AIGR Deep Research Report — *S. pombe* nce101 (C6Y4B6): "non-classical protein export" function

Gene: nce101 / SPAC12G12.17 (SCHPO) · **UniProt:** C6Y4B6 (NCE1_SCHPO) · **Family:** NCE101 (Pfam PF11654 / InterPro IPR024242 / PANTHER PTHR28011) **Seed hypothesis:** *S. pombe* nce101 **directly** participates in signal-sequence-independent non-classical protein export, **rather than merely inheriting** an ambiguous family-level annotation from the *S. cerevisiae* NCE101 screen.

Executive Judgment

Verdict: REFUTED / OVER-ANNOTATED (with respect to "direct participation").

The evidence points to the *opposite* of the seed hypothesis. Far from having direct, gene-product-specific evidence, *S. pombe* nce101 carries **no experimental functional data of any kind**, and every "non-classical export / exocytosis / protein secretion" annotation is electronically or phylogenetically transferred. This is precisely the "inherited ambiguous family-level annotation" scenario the hypothesis was framed to exclude.

Key pillars: 1. **No direct evidence in fission yeast.** UniProt C6Y4B6 is a 58-aa single-TM peptide at Protein Existence level 2 (transcript only). Its FUNCTION comment is 100% "By similarity" (ECO:0000250); GO terms are ISO (exocytosis), IBA (protein secretion), and IEA (membrane). 2. **PomBase itself calls it uncharacterized — and formally annotates it "unknown."** Characterisation status = **"conserved unknown"**; product name is the family label *"non-classical export protein family Nce101."* QuickGO confirms PomBase curates all three GO roots as **No Data (ND, GO_REF:0000015): molecular_function (GO:0003674) = ND, biological_process (GO:0008150) = ND, cellular_component (GO:0005575) = ND.** The only non-root functional terms are **GO:0009306 "protein secretion" via IEA (InterPro IPR024242) + IBA (GO_Central, from the *S. cerevisiae* NCE101 node S000003742)** and **GO:0016020 membrane via IEA.** No experimental (EXP/IDA/IMP/IGI) annotation exists. The only phenotypes are generic chemical-genomic drug/stress sensitivities. 3. **The founder annotation**

is weak and ambiguous. *S. cerevisiae* NCE101 (Q02820, PE = 3 "inferred from homology") rests on a single 1996 galectin-1 **over-expression** screen (P 8655575) and is explicitly hedged: *"may be part of the export machinery or may also be a substrate."* 4. **The "NCE" family is a phenotype label, not a molecular-function family.** Sibling screen hits were reassigned to unrelated functions: **NCE103** = **β-carbonic anhydrase** (an enzyme; P 25109265) and **NCE102** = **plasma-membrane sphingolipid sensor / eisosome-MCC microdomain protein** (P 35758748) — neither a dedicated export-machinery component.

1. **Both reference databases call it "unknown function."** SGD's curated description of the founder (Verified ORF YJL205C) is *"Protein of unknown function; … SWAT-GFP and mCherry fusion proteins localize to the cytosol"* — and PomBase calls the *S. pombe* gene "conserved unknown." The experimental cytosolic localization further **conflicts** with the transferred "single-pass membrane protein" CC (a TMHMM prediction), so even the location term is uncertain.

Most important caveat: "Refuted" here means the *direct-participation* claim is unsupported and the annotation is over-strong; it does **not** prove nce101 is uninvolved in secretion. It is a genuine, fungi-conserved small peptide of currently unknown molecular function (Verified ORF, not dubious). Absence of evidence ≠ evidence of a different function. Note the location is itself ambiguous: predicted single-TM vs experimentally cytosolic tag fusions.

Evidence Matrix

#	Citation	Evidence type	Stance	Claim tested	Key finding	Cont
1	UniProt C6Y4B6 (NCE1_SCHPO)	Database record	Qualifies → refutes	Does *S. pombe* nce101 have direct export evidence?	58-aa single-pass TM peptide; PE = 2 (transcript only); FUNCTION entirely "By similarity" (ECO:0000250); GO exocytosis = ISO, protein secretion = IBA, membrane = IEA	*S. pon*
2	PomBase SPAC12G12.17	Database record	Refutes	Is nce101 experimentally characterized?	**characterisation_status = "conserved unknown"**; product = "non-classical export protein family Nce101"; no experimental GO MF/BP/CC; deletion_viability = unknown; taxonomic distribution = fungi only	*S. pon*
3	PomBase phenotype set (FYPO)	Mutant phenotype (HTP)	Qualifies (not export-specific)	Do deletion phenotypes indicate export role?	Only generic chemical-genomic hits: resistance/sensitivity to amorolfine, EGTA, Li^{+}, SDS combinations, MMS, cadmium, diamide, valproate, vanadate	*S. pon* genor wide scree
4	UniProt Q02820 (NCE1_YEAST)	Database record	Qualifies	Strength of founder annotation	53-aa peptide; PE = 3 "inferred from homology"; GO protein secretion = IGI	*S. cerev*
5	Cleves, Cooper, Barondes, Kelly 1996 —	Direct assay (screen)	Competing / qualifies	Origin of "non-classical export" label	Screen using heterologous mammalian galectin-1 over-expression	*S. cerev*

#	Citation	Evidence type	Stance	Claim tested	Key finding	Cont
	PMID **8655575** (DOI 10.1083/ jcb.133.5.1017)				identified NCE genes; UniProt function is hedged "machinery **or** substrate"	galec repor
6	Zahumenský et al. 2022 — PMID **35758748**	Localization / mechanism	Competing (family reassignment)	Is the "NCE" family a molecular-function family?	NCE102 is a plasma-membrane **sphingolipid sensor** redistributing in eisosome/MCC microdomains	*S. cerev*
7	Lehneck & Pöggeler 2014 — PMID **25109265**	Structural/ enzymatic review	Competing (family reassignment)	Same as #6	NCE103 is a structurally characterized **β-carbonic anhydrase** (enzyme), unrelated to export	Fung CAs i *S. cerev* Nce1
8	UniProt PF11654 family listing (50+ entries)	Computational/ evolutionary	Qualifies	Is there any experimental anchor in the family?	All NCE101-family members are PE 3 (inferred) or PE 4 (predicted); fungi-only; no experimental characterization anywhere	Pan- funga
9	SGD locus YJL205C/ NCE101	Localization / database	Refutes / qualifies	Is the founder characterized, and where does it localize?	Verified ORF but **"Protein of unknown function"**; SWAT-GFP and mCherry fusions localize to the **cytosol** (conflicts with predicted single-pass-membrane CC)	*S. cerev*

#	Citation	Evidence type	Stance	Claim tested	Key finding	Cont
10	UniProt Q12207 (NCE2_YEAST) vs Q02820/ C6Y4B6	Structural/ evolutionary	Qualifies	Paralog confusion between NCE101 and NCE102?	NCE101 = Pfam **PF11654** (53–58 aa, 1 predicted TM); NCE102 = **PF01284** MARVEL/tetraspanin (173 aa, multi-TM) — unrelated families	Cross speci
11	QuickGO annotation set for C6Y4B6	Database (provenance)	Refutes	What does PomBase actually assert for nce101?	PomBase curates **MF, BP, and CC all** = **ND** (No Data, GO_REF:0000015); "protein secretion" exists **only** as IEA (InterPro) + IBA (from *Sc* NCE101 node S000003742); membrane = IEA; **no experimental annotation**	*S. po*

GO Curation Implications *(leads — require curator verification)*

The three functional GO annotations on C6Y4B6 all trace, ultimately, to transferred/ambiguous evidence:

GO term	Aspect	Current evidence	Lead
GO:0009306 protein secretion	BP	IBA (GO_Central) + IEA (InterPro)	**Weaken / flag as non-core.** Supported only electronically + phylogenetically from the ambiguous *Sc* founder node; PomBase itself curates BP root as ND (unknown). Retain at most as low-confidence IBA, or defer to the ND ("unknown biological process") position.
GO:0003674 / GO:0008150 / GO:0005575 (roots)	MF/BP/CC	ND (PomBase, GO_REF:0000015)	**Respect the ND curation.** PomBase explicitly records molecular function, biological process, and cellular component as *unknown*. Do **not** add any MF export/transporter term.
GO:0006887 exocytosis	BP	ISO (from *S. cerevisiae*)	**Candidate for removal or generalization.** ISO transfer from a founder whose own annotation is "machinery or substrate"; "exocytosis" is more specific than the founder evidence supports.
GO:0016020 membrane	CC	IEA + a real TMHMM helix (aa 10–27)	**Retain only cautiously / flag.** A predicted single-TM helix supports it, but the *S. cerevisiae* ortholog's SWAT-GFP/mCherry fusions localize to the **cytosol** (SGD), a direct conflict. Do not upgrade to a specific membrane system; consider "cytoplasm" as a competing CC pending organism-specific data.

- **No MF term is justified** by current evidence beyond, at most, generic membrane localization. Do **not** assert an export-machinery molecular activity.
- Recommend the curated position mirror PomBase's **"conserved unknown"**: a fungi-conserved small single-pass membrane protein of **unknown molecular function**, with any secretion/export terms explicitly marked as homology-transferred and non-core.
- Avoid defaulting to "protein binding" — no interaction evidence exists for the *S. pombe* product.

Mechanistic Scope

Immediate molecular function being tested: whether the 58-aa nce101 peptide is itself a component (or dedicated substrate) of a signal-sequence-independent export apparatus.

- **Direct gene-product activity:** unknown. The only hard biochemical/structural fact is a single predicted transmembrane helix (aa 10–27) → integral membrane peptide.
- **Downstream / inferred:** the "export" descriptor derives from an *over-expression* phenotype on a heterologous reporter (galectin-1) in a different species (*S. cerevisiae*), then propagated by orthology. This is a pathway-level phenotype, not a demonstrated molecular activity of the endogenous peptide.
- The *S. pombe* deletion phenotypes (drug/stress sensitivities) are pleiotropic and do not localize the protein to an export step.

Conflicts and Alternatives

- **Family carry-over / mis-labeling (strongest alternative):** NCE103 (carbonic anhydrase) and NCE102 (sphingolipid sensor) demonstrate that "NCE" screen membership routinely does **not** correspond to a direct export function. nce101 is the least-characterized member.
- **"Substrate vs machinery" ambiguity:** even the founder UniProt statement cannot decide whether the peptide is machinery or cargo — a fundamental unresolved distinction that undercuts any "directly participates in the export machinery" claim.
- **Species differences:** all *S. pombe* evidence is transferred; there is no fission-yeast export assay for nce101.
- **Over-expression artifact:** the founding phenotype is an over-expression/heterologous-reporter effect, a known source of gain-of-function artifacts that need not reflect endogenous function.

Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
No endogenous localization for *S. pombe* nce101	UniProt CC = ECO:0000305; PomBase = conserved unknown	CC term (PM vs internal membrane) constrains plausible function	Endogenous tagged live-cell imaging in *S. pombe*
Machinery vs substrate unresolved	Founder function comment is hedged	Determines whether any MF/BP export term is even appropriate	Interactome + secretome comparison of Δnce101
No secretion assay in fission yeast	PomBase phenotypes are drug/stress only	Direct test of export role is absent	Leaderless-cargo (e.g., Acb1/GAPDH-type) secretion assay in Δnce101
Family function anywhere	Whole PF11654 = PE 3/4	If any ortholog were characterized it would anchor the family	Targeted biochemistry on any tractable ortholog

Discriminating Tests

1. **Endogenous localization:** C-terminal tag of *S. pombe* nce101 at native locus; ask PM/eisosome vs ER/Golgi vs cytosolic membranes.
2. **Leaderless-secretion functional test:** measure unconventional secretion of an established cargo (e.g., Acb1, or heterologous galectin-1/GAPDH) in wild-type vs Δnce101 and vs over-expression — replicating the founder assay in the *actual* organism.
3. **Interactome:** AP-MS / proximity labeling to test physical association with any secretion apparatus vs membrane-microdomain proteins (parallels the NCE102 outcome).
4. **AlphaFold/structural check:** compare the modeled fold/topology against NCE102 (tetraspanin-like) to assess whether nce101 shares a microdomain-protein architecture rather than a transporter fold.
5. **Cross-species annotation audit:** confirm no experimental anchor exists anywhere in PF11654 before propagating export terms.

Curation Leads *(all require curator verification)*

- **Action change:** Treat the "non-classical protein export / exocytosis" function as **non-core, homology-transferred, and over-strong** for C6Y4B6. Align the review with PomBase's **"conserved unknown"** characterisation.
- **GO leads:**
- Retain **GO:0016020 membrane (CC, IEA)** — supported by a predicted TM helix (aa 10–27).
- Flag/weaken **GO:0006887 exocytosis (ISO)** and **GO:0009306 protein secretion (IBA)** as non-experimental transfers; consider removing the more specific "exocytosis" term. Do **not** add an MF export/transporter term.
- **Candidate references to cite for the review:**
- PMID **8655575** (founder screen; snippet: *"A screen for components of the export machinery has identified genes that are involved in nonclassical export."*) — note it is a heterologous over-expression screen and the UniProt function is hedged "machinery or substrate."
- PMID **35758748** (NCE102 = sphingolipid sensor) and PMID **25109265** (NCE103 = carbonic anhydrase) — precedents that "NCE" ≠ molecular export function.
- **Suggested curator questions:**
- Is there *any* experimental (non-IBA/ISO/IEA) evidence for *S. pombe* nce101 function? (Current answer: no.)
- Should family-name-derived BP terms be demoted to "unknown molecular function" pending an organism-specific assay?
- **Suggested experiment (highest value):** Δnce101 leaderless-cargo secretion assay in *S. pombe* (Discriminating Test #2).

Provenance

All programmatic results above were retrieved live during this run via the UniProt REST API (C6Y4B6, Q02820, Q12207 NCE102, PF11654 family search), the PomBase gene API (SPAC12G12.17), and the SGD backend locus API (YJL205C), plus PubMed abstracts (PMIDs 8655575, 35758748, 25109265). No local repository bioinformatics files were used. No result was fabricated; where evidence is absent it is stated as absent.

Iteration 2 additions: SGD confirms YJL205C is a *Verified* (not dubious) ORF but "protein of unknown function" with **cytosolic** GFP/mCherry localization; NCE101 (PF11654) and NCE102 (PF01284, MARVEL) are confirmed unrelated families, ruling out paralog confusion.

Iteration 3 additions: QuickGO API confirms PomBase curates C6Y4B6's MF/BP/CC roots as **ND** (unknown; GO_REF:0000015) and that "protein secretion" is present only as IEA + IBA transfers (IBA source = *S. cerevisiae* NCE101 node S000003742) — the definitive database-level demonstration that the export annotation is inherited, not direct. Computed provenance saved as `artifacts/nce101_evidence_matrix.csv` and `artifacts/nce101_go_decision_table.csv`.
